ME 333

MECHANICS OF MATERIALS

Lecture 22: Introduction to Formulation of Linear Elasticity Problems

Ravi Sastri Ayyagari

Mechanical Engineering
Indian Institute of Technology Gandhinagar

Sem I, 2025-2026

General Relations in 3D

- There are 3 displacements, 6 strains, 6 stresses that are unknowns.
- We require 15 equations to uniquely solve for these unknowns for a body subjected to boundary loads and/or displacements.
- Strain - displacement relations (6) :

$$\epsilon_{xx} = \frac{\partial u}{\partial x}; \quad \epsilon_{xy} = \frac{1}{2}\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right);$$
$$\epsilon_{yy} = \frac{\partial v}{\partial y}; \quad \epsilon_{yz} = \frac{1}{2}\left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right);$$
$$\epsilon_{zz} = \frac{\partial w}{\partial z}; \quad \epsilon_{zx} = \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)$$

General Relations in 3D

- Equations of Equilibrium (3) :

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} = 0$$
$$\frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} = 0$$
$$\frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} = 0$$

- Stress - strain relations (6) :

$$\sigma_{xx} = (\lambda + 2\mu)\epsilon_{xx} + \lambda(\epsilon_{yy} + \epsilon_{zz})$$
$$\epsilon_{xx} = \frac{\lambda + G}{G(3\lambda + 2G)}[\sigma_{xx} - \frac{\lambda}{2(\lambda + G)}(\sigma_{yy} + \sigma_{zz})]$$
$$= \frac{1}{E}[\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})]$$

General Relations in 3D

- Compatibility relations (6) :

$$\epsilon_{xx,yy} + \epsilon_{yy,xx} = 2\epsilon_{xy,xy}$$
$$\epsilon_{yy,zz} + \epsilon_{zz,yy} = 2\epsilon_{yz,yz}$$
$$\epsilon_{zz,xx} + \epsilon_{xx,zz} = 2\epsilon_{xz,xz}$$
$$\epsilon_{zz,xy} + \epsilon_{xy,zz} = \epsilon_{yz,zx} + \epsilon_{zx,yz}$$
$$\epsilon_{yy,xz} + \epsilon_{xz,yy} = \epsilon_{yz,yx} + \epsilon_{xy,yz}$$
$$\epsilon_{xx,yz} + \epsilon_{yz,xx} = \epsilon_{xz,xy} + \epsilon_{xy,xz}$$

- Note: The compatibility equations are used only to ascertain the uniqueness and permissibility of the displacement fields.

Working Assignment

Work out the plane stress and plane strain equations